

Md. Ghazanfar Alam

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Associate Engineer – Emerging Technologies

Associate Engineer specializing in **Generative AI (Gen AI) and Agentic AI**, with strong expertise in **cloud-based AI deployments, LLM frameworks, and AI workflow orchestration**. Certified in **Google Cloud Professional ML Engineer, Data Engineer, Associate Cloud Engineer, and Cloud Digital Leader**. Completed **Gen AI GCP L300 & L400 courses** and 17+ weeks of **intensive hands-on training** in Generative AI, Agentic AI frameworks, and production-ready AI solutions.

Hands-on experience with **LangChain, LangGraph, LlamaIndex, CrewAI, Google ADK, Agno, LangFlow, Flowise**, and **vector stores (FAISS, Pinecone)** for building RAG pipelines, chatbots, and multi-agent AI systems. Skilled in **prompt engineering, multi-agent orchestration, evaluation with OpenEval & DeepEval**, and **deployment on cloud platforms**.

Passionate about leveraging **Gen AI to solve real-world problems**, build scalable AI applications, and implement production-ready AI systems.

Core Competencies

- Gen AI development & LLM Integration
- Computer Vision
- AI Agents
- Deep Learning
- RAG, Retrieval Optimization & Indexing
- Prompt Engineering
- Model Fine Tuning, Optimization & Evaluation
- Vector Databases
- Multi Agent Systems

Tools and Technologies

- Python
- LangGraph
- MySQL
- OpenAI
- Atlassian (Jira, Confluence)
- Google ADK
- Git/Github
- CrewAI
- AgenticAI
- GCP
- LangSmith
- React
- Streamlit
- Llama Index

Responsibilities

- Support strategic planning and execution to meet organizational objectives, ensuring alignment with project goals.
- Design, build, test, and maintain robust software components by applying core programming and machine learning principles.
- Develop and implement generative AI and agentic AI solutions, leveraging frameworks like LangChain, LangGraph, and LlamaIndex.
- Collaborate effectively in dynamic environments, demonstrating adaptability in handling shifting project priorities and complex technical challenges.
- Continuously optimize AI workflows and deployment processes on cloud platforms, ensuring scalable and production-ready applications.

Educational Qualifications

- B.Tech (Computer Science & Engineering)– Sister Nivedita University (2019–2023)

Achievements & Certifications

- Google Cloud Professional Machine Learning Engineer
- Google Cloud Associate Cloud Engineer
- Google Cloud Generative AI Leader
- Google Cloud Digital Leader
- Virtusa Certified Gen AI Assisted Engineer
- Virtusa Jatayu Hackathon Season 2 – Runner-Up

Google Project in Virtusa:

Impact Based Test Selection - Gmail	
Role	Software Developer / AI Engineer
Technology	Python, Google Agent Development Kit(ADK), AgentTools, , Pydantic, JSON, Google(Internal Tools), Cloud Spanner,
Description	● Built an AI-powered Impact-Based Test Selection Agent to automate identification, prioritization, and recommendation of relevant test cases based on Change Lists (CLs) and manual test requests. ● Designed and developed a hybrid multi-agent orchestration pipeline integrating Gemini Enterprise LLMs, Cloud Spanner, and Google internal platforms including Critique, Buganizer, and Test Tracker. ● Engineered intelligent sub-agents for CL verification, test request validation, deduplication, and impact analysis to eliminate redundant testing workflows and improve QA efficiency. ● Developed an advanced deduplication engine using Infra Spanner embeddings and hybrid similarity search to identify previously processed requests and reduce repetitive analysis. ● Implemented automated validation logic for manual test requests, including mandatory field verification, link accessibility checks, and automated feedback generation via comments. ● Integrated historical execution data, bug metadata, code diffs, and feature-level insights to calculate risk scores and determine optimal test coverage requirements. ● Built intelligent recommendation categorization logic to classify tests into Must Run, Good to Have, and Exploratory tiers for prioritized execution planning. ● Optimized orchestration pipelines for scalable, production-grade performance and reliability, significantly reducing regression test analysis time and improving recommendation accuracy. ● Reduced regression test request analysis time by 96% (from 2 hours to 5 mins) and CL analysis time by 75%, saving an estimated 358+ engineering hours annually while maintaining >70% recommendation accuracy.

Impact Based Test Selection - Chat	
Role	Software Developer / AI Engineer
Technology	Python, Google Agent Development Kit(ADK), AgentTools, , Pydantic, JSON, Google(Internal Tools), Cloud Spanner,
Description	● Developed an AI-powered Impact-Based Test Selection Agent to automate identification, prioritization, and recommendation of relevant test cases based on Change Lists (CLs) and manual test requests. ● Designed a hybrid multi-agent orchestration pipeline integrating Gemini Enterprise LLMs, Cloud Spanner, and Google internal engineering platforms including Critique, Bugarizer, and Test Tracker. ● Engineered intelligent AI sub-agents for: ○ CL Verification ○ Test Request Validation ○ Deduplication ○ Impact Analysis ● Developed an advanced deduplication engine using Cloud Spanner embeddings and hybrid similarity search to identify previously processed requests and eliminate redundant analysis workflows. ● Implemented automated validation mechanisms for manual test requests, including mandatory field verification, link accessibility checks, relevancy analysis, and automated feedback generation. ● Integrated historical execution data, bug metadata, feature-level insights, and code diffs to calculate risk scores and determine optimal test coverage requirements. ● Built intelligent recommendation categorization logic to classify tests into: ○ Must Run – Critical high-risk tests ○ Good to Have – Broader related coverage ○ Exploratory – Optional tests for complex scenarios ● Optimized orchestration workflows for scalable, production-grade reliability and high-performance execution.

Training Project in Virtusa:

RMs_AgenticAI – Multi-Agent Financial Analysis System (LangGraph, Gen AI, ML,)	
Role	Software Developer / AI Engineer
Technology	Python, LangGraph, Google Gemini, Scikit-learn, Streamlit,
Description	Designed and developed an end-to-end multi-agent autonomous financial analysis system for Relationship Managers using LangGraph, LLMs, and Machine Learning models. Implemented a production-grade 5-stage workflow orchestrating 10+ specialized AI agents to automate prospect evaluation, investment suitability, compliance checks, and meeting preparation
Responsibilities	• Built a 5-step LangGraph workflow coordinating 10+ agents for automated prospect analysis. • Implemented validated Critical/Optional agent flows with fallback logic and error handling. • Developed ML-driven Risk Profiling and Goal Success Prediction with rule-based backup. • Created a Product Recommendation Engine and Portfolio Optimizer using suitability and risk alignment. • Integrated Google Gemini for persona insights, risk explanations, and RM chat assistance. • Built a Streamlit dashboard with analytics, KPIs, and workflow performance monitoring. • Delivered a production-ready system that automates data validation, financial assessment, and product recommendation processes.

Impact & Outcome	• Automated 70–80% of manual RM workload with faster, accurate, and personalized financial insights. • Delivered a modular, scalable, and extensible multi-agent architecture with real-time explainability. • Demonstrated expertise in Agentic AI, LangGraph orchestration, ML + LLM integration, and production of AI engineering.
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RAG Based PDF Chatbot	
Technology	Python, LangChain, RAG (Retrieval Augmented Generation), React, LLM Integration, Google Gemini, Rest API, Prompt Engineering

Description	• Developed an interactive chatbot capable of answering questions from uploaded PDF documents using RAG and Generative AI. • Built a Flask backend to handle PDF uploads and retrieval-based question answering. • Create a React frontend for real-time chatbot interaction, including chat history and upload validation. • Integrated LangChain for efficient document retrieval and QA workflows. • Implemented REST API communication between frontend and backend for seamless data flow. • Gained hands-on experience in end-to-end AI application development, including deployment of RAG-based systems.
Responsibilities	• As this project was for my own learning, I worked on the whole project.

Handyman Services At your Doorsteps	
Technology	React Native, Typescript, Node.js, MongoDB
Description	In the "SmartService" project, we created two interconnected ReactNative mobile apps: one for customers to book services and one for handymen to accept jobs in real time via sockets. An admin panel, also built in React, ensures efficient oversight. The system was designed to be highly cloud-ready and resilient.

Responsibilities	• Developed and implemented back-end and front-end features using Typescript and React Native. • Analyzed and understood specification requirements to ensure accurate implementation. • Participated in technical development, coding, implementation, and testing processes. • Collaborated effectively within a team of 5 members to achieve project goals. Engaged in code reviews to uphold high standards of code quality and ensure adherence to best. • Frontend - Backend development with API integration
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